Mathematical Structure of Quantum Mechanics (I) Problem Set 3

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10

1. $f(\lambda)=e^{\lambda A}e^{\lambda B}e^{-\lambda(A+B)}$

$$\Rightarrow \frac{df}{d\lambda}=Ae^{\lambda A}e^{\lambda B}e^{-\lambda(A+B)}+e^{\lambda A}Be^{\lambda B}e^{-\lambda(A+B)}+e^{\lambda A}e^{\lambda B}(-A-B)e^{-\lambda(A+B)}$$

$$=Ae^{\lambda A}e^{\lambda B}e^{-\lambda(A+B)}+e^{\lambda A}Be^{\lambda B}e^{-\lambda(A+B)}-e^{\lambda A}e^{\lambda B}Ae^{-\lambda(A+B)}-e^{\lambda A}e^{\lambda B}Be^{-\lambda(A+B)}$$

$$\left(\because Be^{\lambda B}=e^{\lambda B}B\right)$$

$$=Ae^{\lambda A}e^{\lambda B}e^{-\lambda(A+B)}-e^{\lambda A}e^{\lambda B}Ae^{-\lambda(A+B)}$$

$\because$ $e^{\lambda A}e^{\lambda B}Ae^{-\lambda(A+B)}$

$$=e^{\lambda A}\underline{e^{\lambda B}Ae^{-\lambda B}}e^{\lambda B}e^{-\lambda(A+B)}$$

$$=e^{\lambda A}\Big(A+\lambda[B,A]+\frac{\lambda^2}{2}[B,[B,A]]+\cdots\Big)e^{\lambda B}e^{-\lambda(A+B)}$$

($[B,A]=0$, $[B,[B,A]]=0$)

$$=Ae^{\lambda A}+\lambda e^{\lambda A}[B,A]e^{\lambda B}e^{-\lambda(A+B)}$$

$\therefore$ $\frac{df}{d\lambda}=\lambda e^{\lambda A}[A,B]e^{\lambda B}e^{-\lambda(A+B)}$

$$=\lambda e^{\lambda A}[A,B]\underline{e^{-\lambda A}e^{\lambda A}}e^{\lambda B}e^{-\lambda(A+B)}$$

$$=\lambda\Big([A,B]+\lambda[A,[A,B]]+\frac{\lambda^2}{2!}[A,[A,[A,B]]]+\cdots\Big)f(\lambda)$$

($[A,[A,B]]=0$, $[A,[A,[A,B]]]=0$)

$$=\lambda[A,B]f(\lambda)$$

By integrating both sides, $f(\lambda)=\exp\left[\frac{\lambda^2}{2}[A,B]\right]$,

let $\lambda=1$, then $e^{A}e^{B}e^{-(A+B)}=e^{\frac{1}{2}[A,B]}$

$$\Rightarrow e^{A}e^{B}=e^{\frac{1}{2}[A,B]}e^{A+B}$$

$\because\ [\mathbb{A}+\mathbb{B},[\mathbb{A},\mathbb{B}]]=[\mathbb{A},[\mathbb{A},\mathbb{B}]]+[\mathbb{B},[\mathbb{A},\mathbb{B}]]=0+0=0$

$\Rightarrow \mathbb{A}+\mathbb{B}$ and $[\mathbb{A},\mathbb{B}]$ are commute.

$\therefore\ e^{\mathbb{A}}e^{\mathbb{B}}=e^{\frac{1}{2}[\mathbb{A},\mathbb{B}]}e^{\mathbb{A}+\mathbb{B}}=e^{\mathbb{A}+\mathbb{B}+\frac{1}{2}[\mathbb{A},\mathbb{B}]}$ ※

16

3. $\langle\psi_s|[Q,[Q,H]]|\psi_s\rangle$

$=\langle\psi_s|[Q,QH-HQ]|\psi_s\rangle$

$=\langle\psi_s|(QQH-2QHQ+HQQ)|\psi_s\rangle$

* (如果助教覺得不能用Dirac notation，最後一頁有內積的寫法，這頁只是太順手就用了Dirac notation)

$\langle\psi_s|QQH|\psi_s\rangle=\langle\psi_s|QQE_s|\psi_s\rangle=E_s\langle\psi_s|QQ|\psi_s\rangle=$

$=\sum_k E_s\langle\psi_s|Q|\psi_k\rangle\langle\psi_k|Q|\psi_s\rangle$

$=\sum_k E_s|Q_{sk}|^2$.

$\langle\psi_s|HQQ|\psi_s\rangle=E_s\langle\psi_s|QQ|\psi_s\rangle=\sum_k E_s|Q_{sk}|^2$

$\langle\psi_s|QHQ|\psi_s\rangle=\sum_i\sum_j\langle\psi_s|Q|\psi_i\rangle\langle\psi_i|H|\psi_j\rangle\langle\psi_j|Q|\psi_s\rangle$

$=\sum_i\sum_j Q_{is}\cdot E_j\cdot\delta_{ij}\cdot Q_{sj}=\sum_j E_j|Q_{sj}|^2=\sum_k E_k|Q_{sk}|^2$

$\therefore\ \langle\psi_s|[Q,[Q,H]]|\psi_s\rangle=2\left(\sum_k(E_s-E_k)|Q_{sk}|^2\right)$

$[Q,[Q,H]]=[Q,[Q,\frac{p^2}{2m}+V(Q)]]=[Q,\frac{1}{2m}[Q,p^2]]$

$=[Q,\frac{1}{2m}\cdot 2\cdot i\hbar p]=i\hbar\cdot\frac{1}{m}[Q,p]=-\frac{\hbar^2}{m}$

$\Rightarrow\ 2\sum_k(E_s-E_k)|Q_{sk}|^2=\langle\psi_s|[Q,[Q,H]]|\psi_s\rangle=-\hbar^2/m$

$\Rightarrow\ \sum_k(E_k-E_s)|Q_{sk}|^2=\frac{\hbar^2}{2m}$ ※

4. $\because e^{\lambda A} B e^{-\lambda A} = B + \lambda[A,B] + \frac{\lambda^2}{2!}[A,[A,B]] + \frac{\lambda^3}{3!}[A,[A,[A,B]]]$

10 $+ \frac{\lambda^4}{4!}[A,[A,[A,[A,B]]]] + \cdots$

Let $\lambda = \frac{i}{\hbar}\theta$, $A = xP_y - yP_x$, $B = x$,

then $[A,B] = [xP_y - yP_x, x] = [xP_y, x] - [yP_x, x]$

$= x[P_y, x] + [x,x]P_y - y[P_x, x] - [y,x]P_x$

$= i\hbar y$.

$[A,[A,B]] = [xP_y - yP_x, i\hbar y]$

$= i\hbar([xP_y, y] - [yP_x, y]) = i\hbar \cdot (-i\hbar x)$

$[A,[A,[A,B]]] = -(i\hbar)^2[xP_y - yP_x, x] = -(i\hbar)^3 y$

$\vdots$

$\therefore \exp^{\frac{i\theta}{\hbar}(xP_y - yP_x)} x\, e^{-\frac{i\theta}{\hbar}(xP_y - yP_x)}$

$= x + \left(\frac{i\theta}{\hbar}\right)\cdot(i\hbar y) + \frac{1}{2!}\left(\frac{i\theta}{\hbar}\right)^2 \cdot (-)(i\hbar)^2 x + \frac{1}{3!}\left(\frac{i\theta}{\hbar}\right)^3 \cdot (-)(i\hbar)^3 y + \cdots$

$= x - \theta y - \frac{1}{2!}\theta^2 x + \frac{1}{3!}\theta^3 y + \cdots$

$= x\left(1 - \frac{\theta^2}{2!} + \cdots\right) - y\left(\theta - \frac{1}{3!}\theta^3 + \cdots\right) = x\cos\theta - y\sin\theta$ //.

Similiarly, let $C = y$.

then $[A,C] = [xP_y - yP_x, y] = -i\hbar x$

$[A,[A,C]] = (-i\hbar)[A,x] = -(i\hbar)^2 y$

$[A,[A,[A,C]]] = (i\hbar)^3 x$

$$\therefore e^{\frac{i}{\hbar}\theta(xp_y - yp_x)}\, y\, e^{-\frac{i}{\hbar}\theta(xp_y - yp_x)}$$

$$= y + \left(\frac{i\theta}{\hbar}\right)(-i\hbar)x + \frac{1}{2!}\left(\frac{i\theta}{\hbar}\right)^2(-)(i\hbar)^2 y + \frac{1}{3!}\left(\frac{i\theta}{\hbar}\right)^3(i\hbar)^3 x + \cdots$$

$$= y\left(1 - \frac{\theta^2}{2!} + \cdots\right) + x\left(\theta - \frac{\theta^3}{3!} + \cdots\right) = y\cos\theta + x\sin\theta \ \#$$

2. 10

By proposition 7 in the text book, there exists a complete set of states which are simultaneously eigenvectors of A_1 and H.

Let ψ_{α,a_1} be these eigenvectors such that $\begin{cases} A_1\psi_{\alpha,a_1} = a_1\psi_{\alpha,a_2} \\ H\psi_{\alpha,a_1} = E_\alpha\cdot\psi_{\alpha,a_1} \end{cases}$.

Similiarly, there exists simultaneously eigenvectors of A_2 and H.

Let ϕ_{α,a_2} be these eigenvectors such that $\begin{cases} A_2\phi_{\alpha,a_2} = a_2\phi_{\alpha,a_2} \\ H\phi_{\alpha,a_2} = E_\alpha\phi_{\alpha,a_2} \end{cases}$

If the energy eigenstates are non-degenerate, then for any eigenvalue α of H, there is only one eigenvector corresbonding to it.

So that $\psi_{\alpha,a_1} = C\cdot\phi_{\alpha,a_2}$; C is constant.

$$\therefore A_1A_2\phi_{\alpha,a_2} = a_2A_1\phi_{\alpha,a_2} = a_2\cdot A_1\cdot\frac{1}{C}\cdot\psi_{\alpha,a_1} = a_2a_1\cdot\frac{1}{C}\psi_{\alpha,a_1} = a_2a_1\phi_{\alpha,a_2}$$

$$A_2A_1\phi_{\alpha,a_2} = \frac{1}{C}A_2A_1\psi_{\alpha,a_1} = \frac{a_1}{C}A_2\psi_{\alpha,a_1} = a_1A_2\phi_{\alpha,a_2} = a_1a_2\phi_{\alpha,a_2}$$

$\Rightarrow A_1A_2\phi_\alpha = A_2A_1\phi_\alpha$ for all $\alpha \Rightarrow A_1, A_2$ are compatible #

$\therefore$ Energy eigenstates are degenerate #

5. /0

$$[\theta_x L_x + \theta_y L_y + \theta_z L_z,\ x^2+y^2+z^2]$$

$$= \Big[\sum_{i=1}^{3} Q_i \epsilon_{ijk} r_j P_k,\ \sum_{\ell=1}^{3} r_\ell^2\Big]$$

$$= \sum_{i,\ell=1}^{3} [\theta_i \epsilon_{ijk} r_j P_k,\ r_\ell^2]$$

$$= \sum_{i,\ell=1}^{3} \theta_i \epsilon_{ijk} \left(r_\ell [r_j P_k, r_\ell] + [r_j P_k, r_\ell] r_\ell \right)$$

$$= \sum_{i,\ell=1}^{3} \theta_i \epsilon_{ijk} \left(r_\ell \cdot (-i\hbar)\delta_{k\ell} \cdot r_j + r_j \cdot (-i\hbar)\delta_{k\ell} r_\ell \right)$$

$$= \sum_{i=1}^{3} \theta_i \epsilon_{ijk} \cdot (-i\hbar) \cdot 2 \cdot r_j r_k = 0$$

$$\therefore \left[\frac{i}{\hbar}\vec{\theta}\cdot\vec{L},\ x^2+y^2+z^2\right] = 0$$

$$\Rightarrow \exp\left[\frac{i}{\hbar}\vec{\theta}\cdot\vec{L}\right](x^2+y^2+z^2)\exp\left[-\frac{i}{\hbar}\vec{\theta}\cdot\vec{L}\right]$$

$$= x^2+y^2+z^2 + \underbrace{\left[\frac{i}{\hbar}\vec{\theta}\cdot\vec{L},\ x^2+y^2+z^2\right]}_{\to 0} + \cdots$$

$$= x^2+y^2+z^2 \quad \#$$

3. $$[Q, H] = \left[Q,\ \frac{P^2}{2m} + V(Q)\right] = \frac{i\hbar}{m} P$$

$$\Rightarrow [Q,[Q,H]] = \left[Q,\ \frac{i\hbar}{m} P\right] = \frac{i\hbar}{m} \cdot i\hbar\, \mathbb{1} = -\frac{\hbar^2}{m}\mathbb{1}$$

$$\therefore (\psi_S, [Q,[Q,H]]\psi_S) = -\frac{\hbar^2}{m}(\psi_S, \mathbb{1}\psi_S) = -\frac{\hbar^2}{m}$$

Because $[Q,[Q,H]]=[Q,QH-HQ]$

$$= QQH-2QHQ+HQQ$$

Let $Q\psi_s=\sum_k a_k\psi_k$

$$\Rightarrow (\psi_\ell, Q\psi_s)=\sum_k a_k(\psi_\ell,\psi_k)=a_\ell \Rightarrow a_\ell=Q_{\ell s}$$

$$\therefore (\psi_s, QQ\psi_s)=(Q^+\psi_s, Q\psi_s)=\sum_{\ell,k} Q^*_{\ell s}Q_{ks}(\psi_\ell,\psi_k)$$

$$=\sum_{\ell,k} Q^*_{\ell s}Q_{ks}\cdot\delta_{\ell k}=\sum_k |Q_{ks}|^2$$

$$\Rightarrow (\psi_s, QQH\psi_s)=E_s(\psi_s, QQ\psi_s)=E_s\cdot\sum_k |Q_{ks}|^2$$

$$(\psi_s, QHQ\psi_s)=(Q^+\psi_s, HQ\psi_s)=\sum_\ell\sum_k Q^*_{\ell s}Q_{ks}\cdot(\psi_\ell, H\psi_k)$$

$$=\sum_\ell\sum_k Q^*_{\ell s}Q_{ks}\cdot E_k\cdot\delta_{\ell k}=\sum_k E_k|Q_{ks}|^2$$

$$\Rightarrow -\frac{\hbar^2}{m}=(\psi_s,[Q,[Q,H]]\psi_s)$$

$$=(\psi_s,(QQH-2QHQ+HQQ),\psi_s)$$

$$=\sum_k |Q_{ks}|^2(E_s-E_k)$$

$$\Rightarrow \frac{\hbar^2}{2m}=\sum_k (E_k-E_s)|Q_{ks}|^2 \quad \#$$